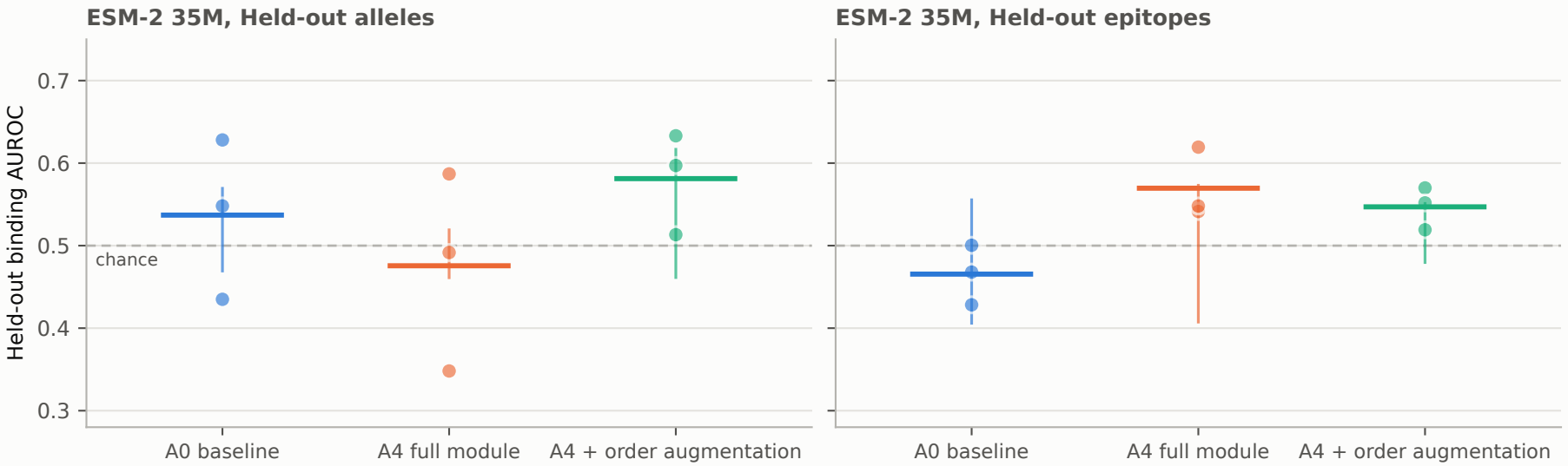
The chain-aware module does not improve held-out binding over a matched baseline

ESM-2 35M. LoRA r=16, trainable parameters matched across arms, 300 steps, three seeds. Each dot is one seed; the bar is the seed mean, the thin line a cluster bootstrap over MHC alleles.



Every arm's interval covers chance and every delta against A0 covers zero. The seed spread is as large as the differences between arms. Negatives are mismatched decoys, so AUROC here is a weak proxy for binding rather than a bound on it.